

量子光学复习.

一、量子力学概要.

① 态: Hilbert 空间中的射线.

② 可观测力学量: 厄米算符 $\hat{A}^\dagger = \hat{A}$: 本征态正交完备, 本征值实数.

③ 测量: $\text{prob} = |\langle \psi | n \rangle|^2$

④ 封闭系统的动力学演化: $i\hbar \partial_t |\psi\rangle = \hat{H} |\psi\rangle$, $|\psi(t)\rangle = \hat{U} |\psi(0)\rangle$. $\hat{H}$ 不含时 $\hat{U} = e^{-i\hat{H}t/\hbar}$

Heisenberg 绘景: $\frac{d\hat{A}}{dt} = \frac{1}{i\hbar} [\hat{A}, \hat{H}]$; $\hat{A} = \hat{U}^\dagger(t) \hat{A}_S \hat{U}(t) = e^{i\hat{H}t/\hbar} \hat{A}_S e^{-i\hat{H}t/\hbar}$

态不演化 $|\psi\rangle_H$

不确定性关系: $[\hat{x}, \hat{p}] = i\hbar$; 态: 概率幅; 本征态正交则可区分, 可重复

谐振子 Fock 态: $\hat{H} = \frac{p^2}{2\mu} + \frac{1}{2} \mu \omega^2 \hat{x}^2 \Rightarrow \begin{cases} \hat{a} = \frac{1}{\sqrt{2\mu\hbar\omega}} (\mu\omega\hat{x} + i\hat{p}) \\ \hat{a}^\dagger = \frac{1}{\sqrt{2\mu\hbar\omega}} (\mu\omega\hat{x} - i\hat{p}) \end{cases} \begin{cases} \hat{a}|n\rangle = \sqrt{n}|n-1\rangle \\ \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \\ \hat{a}^\dagger\hat{a}|n\rangle = n|n\rangle \end{cases}$
 $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + 1/2)$

Fock 态波函数: $\hat{a}|0\rangle = 0$ 可推; $|1\rangle$ 由 $\hat{a}^\dagger|0\rangle = |1\rangle$ 可推

二、辐射场量子化 $\begin{cases} \text{驻波模式 } n\frac{\lambda}{2} = L \Rightarrow \nu_n = \delta\pi c/L. \\ \text{周期性边界条件, 行波模式: } 2\pi/\lambda \cdot L = 2\pi n \Rightarrow L = n\lambda. \end{cases} \quad \nu: \text{光场圆频率}$

Heisenberg 表象得到的随时间演化的算符,

$$\hat{E}(\vec{r}, t) = \sum_{\vec{k}} \hat{e}_{\vec{k}} \sum_{\vec{k}'} \hat{a}_{\vec{k}'} e^{-i\nu_{\vec{k}'}t + i\vec{k}' \cdot \vec{r}} + \sum_{\vec{k}} \hat{e}_{\vec{k}} \sum_{\vec{k}'} \hat{a}_{\vec{k}'}^\dagger e^{i\nu_{\vec{k}'}t - i\vec{k}' \cdot \vec{r}} = \hat{E}^{(+)}(\vec{r}, t) + \hat{E}^{(-)}(\vec{r}, t)$$

态密度: $N_S = 2 \cdot \sum_{\vec{k}} = 2 \cdot \frac{\text{实数}}{(2\pi)^3} \int d^3k$; $\int d^3k = \int dk k^2 \sin\theta d\theta d\varphi = \int d\nu \frac{\nu^2}{c^3} \sin\theta d\theta d\varphi$

以 ν 为自变量的态密度: $D(\nu) = 2 \cdot \left(\frac{L}{2\pi}\right)^3 \frac{\nu^2}{c^3} \cdot 4\pi = \frac{L^3 \nu^2}{\pi^2 c^3}$ 单位频率区间的状态数

• 多模光子数态: $\hat{H} = \sum_{\vec{k}} \hbar\nu_{\vec{k}} (\hat{a}_{\vec{k}}^\dagger \hat{a}_{\vec{k}} + 1/2) \Rightarrow |n_{\vec{k}_1}, n_{\vec{k}_2}, \dots\rangle$ 不同模独立. 注意 $\vec{k}$ 有方向

单模: $\begin{cases} \langle n | \hat{E}^2 | n \rangle = \langle n | \hat{E}^2 \hat{a}^\dagger \hat{a} + \hat{E}^2 \hat{a} \hat{a}^\dagger | n \rangle = |\hat{E}|^2 (n+1); \text{当 } n=0 \text{ 时, } \langle 0 | \hat{E}^2 | 0 \rangle = |\hat{E}|^2 \neq 0 \\ \langle n | \hat{E} | n \rangle = 0 \end{cases}$

• Lamb 移位: 电子受真空电磁场涨落扰动偏离圆周轨道, w/ 哈密顿量在微扰下

$$\langle \Delta V \rangle = \langle \frac{1}{2} (\delta\vec{r} \cdot \vec{p})^2 V(\vec{r}) \rangle = \frac{1}{2} \langle \delta\vec{r}^2 \rangle_{\text{vac}} \langle \nabla^2 \left(-\frac{e^2}{4\pi\epsilon_0 r} \right) \rangle_{\text{atom}}; \nabla^2 \frac{1}{r} = -4\pi \delta(\vec{r})$$

$\frac{\pi}{a_0} < k < \frac{\pi}{a}$ $\downarrow$ 长波整体影响原子 $\downarrow$ 非相对论极限 $\downarrow$ 谐振子半经典解代入 $\downarrow$ S 态波函数代入; 只有 S 态有波函数零非零

Casimir 效应: 板间模式密度小板外模式密度大, 两板被向内挤压

• $[\hat{A}, \hat{B}] \neq 0, [\hat{A}, [\hat{A}, \hat{B}]] = [\hat{B}, [\hat{A}, \hat{B}]] = 0$ 则 $e^{\hat{A}+\hat{B}} = e^{\hat{A}} e^{\hat{B}} e^{-\frac{1}{2}[\hat{A}, \hat{B}]}$

$$\begin{cases} [\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B} \\ [\hat{a}, \hat{a}^{\dagger n}] = n\hat{a}^{\dagger n-1}, [\hat{a}, f(\hat{a}, \hat{a}^\dagger)] = \frac{\partial f}{\partial \hat{a}^\dagger} \\ [\hat{a}^\dagger, \hat{a}^n] = -n\hat{a}^{n-1}, [\hat{a}^\dagger, f(\hat{a}, \hat{a}^\dagger)] = -\frac{\partial f}{\partial \hat{a}} \end{cases}$$

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!} [\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots$$



正规排列: $\hat{a}^\dagger$ 在左, 反正规排列 $\hat{a}$ 在左.

密度算符: 相干态和混合态 $\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$, $\langle\hat{A}\rangle = \text{Tr}(\hat{A}\hat{\rho})$
 $\hat{\rho}^\dagger = \hat{\rho}$, 正定 $\langle\psi|\hat{\rho}|\psi\rangle \geq 0$, $\text{Tr}(\hat{\rho}) = 1$, 纯态 $\Rightarrow \text{Tr}(\hat{\rho}^2) = 1$, 混合态 $\text{Tr}(\hat{\rho}^2) < 1$.

归一化密度: $\hat{\rho}_A = \text{Tr}_B(\hat{\rho}_{AB}) = \sum_m \langle m| \hat{\rho}_{AB} |m\rangle_B$

刘维方程: $\hbar \partial_t \hat{\rho} = [\hat{H}, \hat{\rho}]$ 熵: $S(\hat{\rho}) = -k_B \text{Tr}(\hat{\rho} \ln \hat{\rho})$.

单模热光场 $\hat{\rho} = e^{-\beta \hat{H}} / \mathcal{Z}$, $\mathcal{Z} = \text{Tr}(e^{-\beta \hat{H}})$, $\hat{H} = \hbar \omega (\hat{a}^\dagger \hat{a} + 1/2)$

$\hat{\rho} = (1 - e^{-\beta \hbar \omega}) (\sum_n e^{-n \beta \hbar \omega} |n\rangle\langle n|)$, 多模: $\hat{\rho} = \hat{\rho}_1 \otimes \hat{\rho}_2$

三光场的态与表示: $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$ 相干态 $|\alpha\rangle = \sum_n \frac{e^{-|\alpha|^2/2}}{\sqrt{n!}} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$

$|\alpha\rangle = e^{-|\alpha|^2/2} e^{\alpha \hat{a}^\dagger} e^{-\alpha^* \hat{a}} |0\rangle = e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} |0\rangle = \hat{D}(\alpha) |0\rangle$ ($\hat{D}(\alpha)$ 么正).

(么正性的证明: $e^{\hat{A}}$, 若 $\hat{A}^\dagger = -\hat{A}$, then $e^{\hat{A}^\dagger} e^{\hat{A}} = e^{-\hat{A}} e^{\hat{A}} = \hat{I}$.)

性质: ① 平均光子数: $\langle n \rangle = \langle \hat{a}^\dagger \hat{a} \rangle = |\alpha|^2$.

$p_n = \text{Poi}(\bar{n}) = \langle n | \alpha \rangle^2$ 以 $|\alpha|^2$ 为参数的 Poisson 分布.

$\langle n \rangle$ 大时 $\langle n \rangle$ 附近概率大, $\langle n \rangle < 1$ 时 p_0 最大.

② 不确定性: 最小不确定波包, 正交相位分量: $\hat{x}_1 = \frac{\hat{a} + \hat{a}^\dagger}{2}$, $\hat{x}_2 = \frac{\hat{a} - \hat{a}^\dagger}{2i}$

$\hat{p} = (\hat{a} - \hat{a}^\dagger) \sqrt{\frac{\hbar}{2m\omega}}$; $\hat{p} = \frac{\hat{a} - \hat{a}^\dagger}{i} \sqrt{\frac{\hbar m \omega}{2}}$ $\Delta^2 \hat{x}_1 = \langle \hat{x}_1^2 \rangle - \langle \hat{x}_1 \rangle^2$ $\Delta \hat{x}_1 = \Delta \hat{x}_2 = 1/2$,

$\Delta p \Delta q = \hbar/2 \Leftrightarrow \Delta \hat{x}_1 \Delta \hat{x}_2 = 1/4$

③ 非正交: $\langle \alpha | \alpha' \rangle = e^{-|\alpha|^2/2} e^{-|\alpha'|^2/2} e^{\alpha^* \alpha'}$

④ 超完备: $\int d^2\alpha |\alpha\rangle\langle\alpha| = \pi \hat{I}$ ($d^2\alpha = r dr d\theta$) 复平面的复变积分.

⑤ 相干态自由演化 $i\hbar \partial_t |\psi\rangle = \hat{H} |\psi\rangle$; $|\psi(0)\rangle = |\alpha\rangle \Rightarrow |\psi(t)\rangle = |\alpha e^{-i\omega t}\rangle$.

相干态波函数: $(\hat{a} - \alpha)|\alpha\rangle = 0$, $\tilde{\alpha} = \frac{1}{\sqrt{2\pi\hbar m\omega}} (m\omega x + \hbar \partial_x)$ 可解.

(平移真空态) $\psi(q, t) = (\frac{\nu}{\pi\hbar})^{1/4} \exp(-\frac{\nu}{2\hbar} (q - q_0 e^{-i\omega t})^2)$ 去掉模因子.

$|\psi(q, t)|^2 = (\frac{\nu}{\pi\hbar})^{1/2} \exp(-\frac{\nu}{\hbar} (q - q_0 \cos \omega t)^2)$

以光频为振荡 形状不变在势阱中来回振荡. (最小不确定波包的自由运动)

经典振荡电流驱动量子谐振子, 谐振子从真空态被马吕动作驱动受迫振荡

$\Delta E \Delta t \geq \hbar/2$ 得到 $\Delta n \hbar \omega \Delta t \geq \hbar/2$, $\Delta n \Delta \theta \geq 1/2$; $|n\rangle$ 的 θ 完全不确定, $|\alpha\rangle$ 有 $\Delta n = \sqrt{\text{var}(n)} = \sqrt{\langle n \rangle} = |\alpha|$.

压缩态: 正交相位分量之一, $\Delta^2 \hat{x}_1$ 或 $\Delta^2 \hat{x}_2 < 1/4$ 称压缩态 $|\alpha\rangle, |n\rangle$ 都不压缩

$\hat{S}(\xi) = \exp(\frac{1}{2} \xi^* \hat{a}^2 - \frac{1}{2} \xi \hat{a}^{\dagger 2})$, 偶次算符 $\hat{S}|\xi\rangle$ 向压缩真空态, $\hat{S}|\xi\rangle|\alpha\rangle$ 压缩相干态.

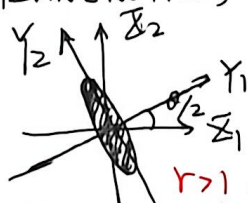
$\xi = r e^{i\theta}$, r 表示压缩程度, θ 压缩方向. $\hat{S}|\xi\rangle$ 为么正算符.

$\hat{S}^\dagger(\xi) \hat{a} \hat{S}(\xi) = \hat{a} \cosh r - \hat{a}^\dagger e^{i\theta} \sinh r \Rightarrow \hat{S}^\dagger \hat{a}^\dagger \hat{S}(\xi) = \hat{a}^\dagger \cosh r - \hat{a} e^{-i\theta} \sinh r$

$\Rightarrow \hat{S}^\dagger \hat{a}^2 \hat{S}$ 补成两个 $= (\hat{a} \cosh r - \hat{a}^\dagger e^{i\theta} \sinh r)^2$



压缩态的生成：压缩相干态 $|\alpha, \zeta\rangle = S(\zeta) D(\alpha) |0\rangle$



$$\begin{cases} \hat{Y}_1 = \cos \frac{\theta}{2} \hat{x}_1 + \sin \frac{\theta}{2} \hat{x}_2 \\ \hat{Y}_2 = -\sin \frac{\theta}{2} \hat{x}_1 + \cos \frac{\theta}{2} \hat{x}_2 \end{cases}$$

$$\Rightarrow \begin{cases} \hat{Y}_1 = \frac{\hat{a} e^{-i\theta/2} + \hat{a}^\dagger e^{i\theta/2}}{2} & \Delta Y_1 = \frac{1}{2} e^{-r} \\ \hat{Y}_2 = \frac{\hat{a} e^{-i\theta/2} - \hat{a}^\dagger e^{i\theta/2}}{2i} & \Delta Y_2 = \frac{1}{2} e^r \end{cases}$$

总之化为 $S^\dagger(\zeta) \hat{a} S(\zeta)$ 就行了： $\hat{S}^\dagger(\zeta) = \hat{S}(-\zeta)$ 可互换身份。 $\Delta Y_1 \Delta Y_2 = 1/4$

先压缩再平移的话， $\langle \hat{x}_1 \rangle = \text{Re} \alpha$ ， $\langle \hat{x}_2 \rangle = \text{Im} \alpha$ ，在相空间以 α 为中心。

• P表示： $\hat{\rho} = \int d^2\alpha p(\alpha) |\alpha\rangle \langle \alpha|$ ($\alpha \in \mathbb{C}$) 相干态： $\hat{\rho}_{\alpha_0} = \int \delta^2(\alpha - \alpha_0) |\alpha\rangle \langle \alpha| d^2\alpha$

求正规式均值： $\text{Tr}(\hat{\rho} f) = \int d^2\alpha \int d^2\beta \langle \alpha | \hat{\rho} | \beta \rangle f(\alpha^*, \beta)$

正规 $\nearrow$ $= \int d^2\alpha p(\alpha) f(\alpha^*, \alpha)$ 反正规。

$$\cdot C_N = \text{Tr}(\rho e^{\lambda \hat{a}^\dagger} e^{-\lambda^* \hat{a}}); C_A = \text{Tr}(\rho e^{-\lambda^* \hat{a}} e^{\lambda \hat{a}^\dagger}); C_W = \text{Tr}(\rho e^{(\lambda \hat{a}^\dagger - \lambda^* \hat{a})})$$

$Q(\alpha) = \frac{1}{\pi} \langle \alpha | \rho | \alpha \rangle \rightarrow$ 求反正规式均值。

$$\begin{cases} p(\alpha) = \frac{1}{\pi} \int d^2\lambda e^{\alpha \lambda^* - \alpha^* \lambda} C_N(\lambda), & C_N(\lambda) = \int d^2\alpha p(\alpha) e^{\lambda \alpha^* - \lambda^* \alpha} \\ Q(\alpha) = \frac{1}{\pi^2} \int d^2\lambda e^{\alpha \lambda^* - \alpha^* \lambda} C_A(\lambda), & C_A(\lambda) = \int d^2\alpha Q(\alpha) e^{\lambda \alpha^* - \lambda^* \alpha} \\ W(\alpha) = \frac{1}{\pi^2} \int d^2\lambda e^{\alpha \lambda^* - \alpha^* \lambda} C_W(\lambda), & C_W(\lambda) = \int d^2\alpha W(\alpha) e^{\lambda \alpha^* - \lambda^* \alpha} \end{cases}$$

• P表示的特点： $\hat{\rho} = \hat{\rho}^\dagger$ ，则 $p(\alpha) = p^*(\alpha)$ 实数； $\text{Tr}(\hat{\rho}) = 1$ 则 $\int d^2\alpha p(\alpha) = 1$ 。

① 准概率分布：非经典态 α ， $p(\alpha) < 0$ ； $p(\alpha) > 0$ 恒成立为经典态 相干态是 $\frac{1}{\pi}$ ！

相干态为边界 或出现奇异。

$$C_W(\lambda) = C_N(\lambda) e^{-|\lambda|^2/2} = C_S(\lambda) e^{|\lambda|^2/2} \quad \text{计算时全化成 } C_A \text{ 的形式然后在中插入一个相干态}$$

可以把特征函数化成 $p(\alpha)$ 或 $Q(\alpha)$ 的形式。

$$= 1 + \frac{\int p(\alpha) (|\alpha|^2 - \langle n \rangle) d^2\alpha}{(\int p(\alpha) |\alpha|^2 d^2\alpha)^2} \quad \text{亚泊松光 } g^{(2)}(0) < 1 \text{ 必然 } p(\alpha) < 0 \rightarrow \langle n \rangle^2$$

• Q表示的特点： $\text{Tr}(\hat{\rho}) = 1$ 则 $\int d^2\alpha Q(\alpha) = 1$ ；

$Q(\alpha) \in [0, \frac{1}{\pi}]$ 非奇异为概率分布。

$$\begin{cases} W(\alpha) = \frac{2}{\pi} \int d^2\beta p(\beta) e^{-2|\beta - \alpha|^2} \\ Q(\alpha) = \frac{1}{\pi} \int d^2\beta p(\beta) e^{-|\beta - \alpha|^2} \end{cases}$$

三、相干性与干涉 概率幅；分束器矩阵描述 物理本质上的不可区分性造成干涉

麦克耳孙测量干涉仪：设两束频率相同：

E_k 和 E_k' 中带有随机初相位

$$I = K \langle |E_k(e^{i\vec{k} \cdot \vec{r}_1} + e^{i\vec{k} \cdot \vec{r}_2}) + E_k'(e^{i\vec{k} \cdot \vec{r}_1} + e^{i\vec{k} \cdot \vec{r}_2})|^2 \rangle = K \langle 2(|E_k|^2 + |E_k'|^2) + (E_k E_k'^* e^{i\vec{k} \cdot (\vec{r}_1 - \vec{r}_2)} + \text{c.c.}) \rangle$$

$$= 4K I_0 (1 + \cos[(\vec{k} + \vec{k}') \cdot (\vec{r}_1 - \vec{r}_2)/2] \cos[(\vec{k} - \vec{k}') \cdot (\vec{r}_1 - \vec{r}_2)/2] + (E_k E_k'^* e^{i\vec{k} \cdot (\vec{r}_1 - \vec{r}_2)} + \text{c.c.}))$$

低频包络： $K \propto \omega/2$ 调 ω_0 测 ω

$$\text{HBT干涉仪: } \langle I(\vec{r}_1, t) I(\vec{r}_2, t) \rangle = K^2 \langle \{ |E_k|^2 + |E_k'|^2 + (E_k E_k'^* e^{i\vec{k} \cdot (\vec{r}_1 - \vec{r}_2)} + \text{c.c.}) \} \{ |E_k|^2 + |E_k'|^2 + (E_k E_k'^* e^{i\vec{k} \cdot (\vec{r}_1 - \vec{r}_2)} + \text{c.c.}) \} \rangle$$

$$= K^2 \langle \{ (|E_k|^2 + |E_k'|^2)^2 + \langle |E_k|^2 \rangle \langle |E_k'|^2 \rangle + 2 \cos(\vec{k} - \vec{k}') \cdot (\vec{r}_1 - \vec{r}_2) \} \rangle \quad \text{只有低频项}$$

光子不是只能和自身干涉：经典干涉无论分波前还是分振幅都是分振幅的单光子干涉
多光子只是增强效应，但 HBT 是强度干涉 不同于时间干涉



一次量化的核心: 谐振子能量和频率 $\hat{E}^{(-)} \propto \hat{a}$, $\hat{E}^{(+)} \propto \hat{a}^\dagger$

光子计数率 $\mathcal{W}_1(\vec{r}, t) = \sum_i p_i \langle i | \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t) | i \rangle = \text{Tr}(\hat{\rho} \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t))$

一阶相关函数: $G^{(1)}(\vec{r}_1, \vec{r}_2; t_1, t_2) = \text{Tr}(\hat{\rho} \hat{E}^{(-)}(\vec{r}_1, t_1) \hat{E}^{(+)}(\vec{r}_2, t_2))$ 有量纲 不相干=不独立
平稳光场: $G^{(1)}(\vec{r}_1, \vec{r}_2; t_1, t_2) = G^{(1)}(\vec{r}_1, \vec{r}_2; t_1 - t_2)$; $\langle \hat{a}^\dagger \hat{a} \rangle$ 平稳, $\langle \hat{a}^\dagger \rangle$ 不一定

光子计数率: $\mathcal{W}_1 = G^{(1)}(\vec{r}, \vec{r}; 0)$, 本地光强; $G^{(1)}(\vec{r}, \vec{r}; 0)$ 本地自相关.

$\mathcal{W}_2(\vec{r}_1, t_1; \vec{r}_2, t_2) = \sum_i | \langle i | \hat{E}^{(+)}(\vec{r}_2, t_2) \hat{E}^{(-)}(\vec{r}_1, t_1) | i \rangle |^2 = \text{Tr}(\hat{\rho} \hat{E}^{(-)}(\vec{r}_1, t_1) \hat{E}^{(+)}(\vec{r}_2, t_2) \hat{E}^{(-)}(\vec{r}_2, t_2) \hat{E}^{(+)}(\vec{r}_1, t_1))$

二阶相关函数 $G^{(2)}(\vec{r}_1, \vec{r}_2, \vec{r}_3, \vec{r}_4; t_1, t_2, t_3, t_4) = \text{Tr}(\hat{\rho} \hat{E}^{(-)}(\vec{r}_1, t_1) \hat{E}^{(-)}(\vec{r}_2, t_2) \hat{E}^{(+)}(\vec{r}_3, t_3) \hat{E}^{(+)}(\vec{r}_4, t_4))$

n阶相关函数 $G^{(n)}(\vec{r}_1, \dots, \vec{r}_n; t_1, \dots, t_n) = \text{Tr}(\hat{\rho} \hat{E}^{(-)}(\vec{r}_1, t_1) \dots \hat{E}^{(-)}(\vec{r}_n, t_n) \hat{E}^{(+)}(\vec{r}_n, t_n) \dots \hat{E}^{(+)}(\vec{r}_1, t_1))$

注: $\langle \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t) \rangle \neq \langle I(\vec{r}, t) I(\vec{r}, t) \rangle$ 不可互换; $\langle I(\vec{r}, t) \rangle = \langle \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t) \rangle$ 光强.

经典相关函数: $\langle E^* E \rangle$, $\langle E^* E^* E E \rangle$, 角标略

相干度: (某位置 $\vec{r}$): $g^{(1)}(\vec{r}, \tau) = \frac{\langle \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t+\tau) \rangle}{\sqrt{\langle \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t) \rangle \langle \hat{E}^{(-)}(\vec{r}, t+\tau) \hat{E}^{(+)}(\vec{r}, t+\tau) \rangle}}$ $0 \leq |g^{(1)}(\vec{r}, \tau)| \leq 1$
 $g^{(2)}(\vec{r}, \tau) = \frac{\langle \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(-)}(\vec{r}, t+\tau) \hat{E}^{(+)}(\vec{r}, t+\tau) \hat{E}^{(+)}(\vec{r}, t) \rangle}{\langle \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t) \rangle \langle \hat{E}^{(-)}(\vec{r}, t+\tau) \hat{E}^{(+)}(\vec{r}, t+\tau) \rangle}$ 不相关 完全相干
 $g^{(2)}(\vec{r}, \tau)$ 允许 > 1

单模场的退化: $g^{(1)}(\tau) = \frac{\langle \hat{a}^\dagger(t) \hat{a}(t+\tau) \rangle}{\langle \hat{a}^\dagger \hat{a} \rangle}$; $g^{(2)}(\tau) = \frac{\langle \hat{a}^\dagger(t) \hat{a}^\dagger(t+\tau) \hat{a}(t+\tau) \hat{a}(t) \rangle}{\langle \hat{a}^\dagger \hat{a} \rangle^2}$

相干态: $|g^{(n)}| = 1, \forall n$

单光子源: $g^{(2)}(0) = 0$. (for 1光子态); $\langle I(\vec{r}, t) \rangle = \langle I^{(1)}(\vec{r}) \rangle + \langle I^{(2)}(\vec{r}) \rangle + 2\sqrt{\langle I^{(1)}(\vec{r}) \rangle \langle I^{(2)}(\vec{r}) \rangle} \text{Re}(g^{(1)}(\vec{r}_1, \vec{r}_2; \tau))$
双光子发光的 Young 干涉: $V = \frac{2\sqrt{\langle I^{(1)}(\vec{r}) \rangle \langle I^{(2)}(\vec{r}) \rangle}}{\langle I^{(1)}(\vec{r}) \rangle + \langle I^{(2)}(\vec{r}) \rangle} |g^{(1)}(\vec{r}_1, \vec{r}_2; \tau)|$
 $g^{(1)}(\vec{r}_1, \vec{r}_2; \tau) = \frac{G^{(1)}(\vec{r}_1, \vec{r}_2; \tau)}{\sqrt{G^{(1)}(\vec{r}_1, \vec{r}_1; 0) G^{(1)}(\vec{r}_2, \vec{r}_2; 0)}}$

Fourier变换: 功率谱 $S(\nu, \nu) = \frac{1}{\pi} \text{Re} \int_0^\infty d\tau G^{(1)}(\vec{r}, \vec{r}; \tau) e^{i\tau\nu}$

二阶实验: HBT在 $|1\rangle = |1\rangle, |2\rangle$ 的二阶相关函数

$G^{(2)}(\vec{r}_1, \vec{r}_2, t_1, t_2) = \langle 1|1\rangle \langle \hat{E}^{(-)}(\vec{r}_1, t_1) \hat{E}^{(-)}(\vec{r}_2, t_2) | 0 \rangle \langle 0 | \hat{E}^{(+)}(\vec{r}_2, t_2) \hat{E}^{(+)}(\vec{r}_1, t_1) | 1\rangle = \psi_2^*(\vec{r}_1, \vec{r}_2; t_1, t_2) \psi_2(\dots)$

同时: $\psi_2(\vec{r}_1, \vec{r}_2, t, t) = \sum_{\vec{k}} \langle 0 | (\hat{a}_{\vec{k}} e^{-i\nu t} e^{i\vec{k} \cdot \vec{r}_2} + \hat{a}_{\vec{k}}^\dagger e^{-i\nu t} e^{i\vec{k} \cdot \vec{r}_2}) (\hat{a}_{\vec{k}} e^{-i\nu t} e^{i\vec{k} \cdot \vec{r}_1} + \hat{a}_{\vec{k}}^\dagger e^{-i\nu t} e^{i\vec{k} \cdot \vec{r}_1}) | 1\rangle \langle 1|$
 $= \sum_{\vec{k}} e^{-2i\nu t} (e^{i\vec{k} \cdot \vec{r}_2 + i\vec{k} \cdot \vec{r}_1} + e^{i\vec{k} \cdot \vec{r}_2 + i\vec{k} \cdot \vec{r}_1})$, $G^{(2)} = |\psi_2|^2 = \dots$

$G^{(2)} = |\psi(\text{X}) + \psi(\text{=})|^2$ 集体行为不可区分.

$G^{(1)} = |\psi(\text{>}) + \psi(\text{<})|^2$ 相位差, 不同路径.

稳定性分析: $\vec{k} \rightarrow \vec{k} + \vec{k}'$ 带来扰动时 $\vec{k} + \vec{k}'$ 项变大(两信扰动值), $\vec{k} - \vec{k}'$ 项相抵扰动不大后者稳定

单模量破磁场所处于Fock和相干态均满足 $|g^{(1)}(x_1, x_2)| = 1$, 区分只是 $n \rightarrow |x|^2$. 无法区分

热光场的计算: $\hat{\rho} = \sum p_n |n\rangle \langle n|$ 因此每一项只有产生和湮灭的次数相同才能保留.

(随布洛散光) $\Rightarrow \hat{E}^{(+)}(\vec{r}_i, t) = \sum_{\vec{k}} (a_{\vec{k}} e^{-i\nu t + i\vec{k} \cdot \vec{r}_i} + a_{\vec{k}}^\dagger e^{-i\nu t + i\vec{k} \cdot \vec{r}_i})$, $G^{(2)} = \dots \frac{P_{123}}{P_{123}}$

Fermi子: $a_{\vec{k}}^\dagger a_{\vec{k}'}^\dagger = -a_{\vec{k}'}^\dagger a_{\vec{k}}^\dagger$; $\langle 0 | a_{\vec{k}} a_{\vec{k}'} | 1\rangle \langle 1| = -1 \Rightarrow \psi_2 \propto e^{-2i\nu t} (e^{i\vec{k} \cdot \vec{r}_2 + i\vec{k} \cdot \vec{r}_1} - e^{i\vec{k}' \cdot \vec{r}_2 + i\vec{k}' \cdot \vec{r}_1})$



反聚束: $g^{(2)}(\tau) > g^{(2)}(0)$ 经典的只有聚束 $g^{(2)}(\tau) \leq g^{(2)}(0)$, 符号为随机

排斥聚束, 某处发现了一个光子更倾向于过一会再发现而非立即发现

泊松: $Var(n) < \langle n \rangle$ 光子数分布方差小, 不确定性低

单模场 $g^{(2)}(0) = g^{(2)}(\tau) = \frac{\langle \hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger \rangle}{\langle \hat{a}^\dagger \hat{a} \rangle^2}$ 单模场不存在反聚束

$$\hookrightarrow = \frac{\langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2}{\langle \hat{n} \rangle^2} = 1 + \frac{Var(n) - \langle \hat{n} \rangle}{\langle \hat{n} \rangle^2} \text{ 可以出现泊松}$$

$$\begin{cases} \text{热场 } g^{(2)}(0) = 2 \text{ 超泊松} \\ \text{相干光 } g^{(2)}(0) = 1 \text{ 泊松} \\ |n\rangle: g^{(2)}(0) = \begin{cases} 0 & 0, 1 \\ 1 - \frac{1}{n} & 2, 3, \dots \end{cases} \end{cases}$$

单模热光场: $\bar{n}$ 和 $\bar{n}^2$ 可求, $Var(n) = \bar{n}(\bar{n}+1)$, $g^{(2)}(0) = 1 + \frac{\bar{n}^2}{\bar{n}^2} = 2$.

光子统计的问题: 探测效率与暗计数

$$n \text{ 个光子探测到 } m \text{ 个: } P = C_n^m \eta^m (1-\eta)^{n-m}; \quad P_m = \sum_n P_{nm} C_n^m \eta^m (1-\eta)^{n-m}$$

零拍探测, 平衡零拍探测

$$\begin{aligned} \hat{c} &= \sqrt{T} \hat{a} + i\sqrt{1-T} \hat{b} \\ \hat{d} &= \sqrt{T} \hat{b} + i\sqrt{1-T} \hat{a} \end{aligned}$$

$$\hat{n}_{cd} = \hat{c}^\dagger \hat{c} - \hat{d}^\dagger \hat{d} \quad \begin{cases} \langle \hat{n}_{cd} \rangle = -2|\beta| \langle \hat{x}(\phi_l + \pi/2) \rangle \\ (\Delta \hat{n}_{cd})^2 = 4|\beta|^2 (\Delta \hat{x})^2 \end{cases}$$

调节本地光的 ϕ_l 则是不同时间的方差, 若某方差 $(\Delta \hat{x})^2 < 1/4$ 表明压缩态.

四. 相互作用半经典理论. $\hat{\sigma}_0 = \hat{I}$, $\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\hat{\sigma}_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\hat{\sigma}_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

哈密量: $\hat{H} = \hat{H}_0 + \hat{H}_I = \hbar \omega_a |a\rangle\langle a| + \hbar \omega_b |b\rangle\langle b| - E(t) (g_{ab} |a\rangle\langle b| + g_{ba} |b\rangle\langle a|)$

$[g_{ab} = e \langle a | \hat{x} | b \rangle]$, 拉比频率 $\Omega_R = \frac{|g_{ab}| E}{\hbar}$, $E(t) = E \cos \omega t$

$|a\rangle$ 上, $|b\rangle$ 下

相互作用绘景: $\hat{A}_I = \hat{U}_0^\dagger \hat{A}_S \hat{U}_0$; 演化(态): $i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{V}_I(t) |\psi(t)\rangle$, $\hat{V}_I = \hat{U}_0^\dagger \hat{V}_S \hat{U}_0$

算符演化: $\frac{d}{dt} \hat{A}_I = \frac{i}{\hbar} [\hat{A}_I, \hat{H}_0]$

$\begin{cases} \tilde{c}_a = c_a e^{i\omega_a t} \\ \tilde{c}_b = c_b e^{i\omega_b t} \end{cases}$ 相当于相互作用绘景变换; 在 Sch 中到 $i\hbar \frac{d}{dt} |\psi\rangle = \hat{H}_I |\psi\rangle$ 即可.

$\Omega = \sqrt{\Delta^2 + \Omega_R^2}$, 失谐 $\Delta = \omega - \omega_0$ 原子激光频率, 布居反转 $w = |c_a|^2 - |c_b|^2$

单模电磁场和单二能级原子的变换: $\hat{H}_0 = \frac{1}{2} \hbar \omega \hat{\sigma}_z$, $\hat{V} = \hbar \Omega_R (e^{-i\omega t} \hat{\sigma}_- + e^{i\omega t} \hat{\sigma}_+)$

$\Rightarrow \hat{U}_0^\dagger \hat{\sigma}_- \hat{U}_0 = \hat{\sigma}_- e^{-i\omega t}$, $\hat{U}_0^\dagger \hat{\sigma}_+ \hat{U}_0 = \hat{\sigma}_+ e^{i\omega t}$

单模光子电磁场和单二能级原子的变换: $\hat{H}_0 = \frac{1}{2} \hbar \omega \hat{\sigma}_z$, $\hat{V} = \hbar g (\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+)$

$\begin{cases} \hat{U}_0^\dagger \hat{\sigma}_- \hat{U}_0 = \hat{\sigma}_- e^{-i\omega t} \\ \hat{U}_0^\dagger \hat{\sigma}_+ \hat{U}_0 = \hat{\sigma}_+ e^{i\omega t} \end{cases}$, $\hat{U}_0^\dagger \hat{a} \hat{U}_0 = \hat{a} e^{-i\omega t}$, $\hat{U}_0^\dagger \hat{a}^\dagger \hat{U}_0 = \hat{a}^\dagger e^{i\omega t}$

特点: 换到相互作用绘景哈密顿量含时, 换到旋波参考架的哈密顿量不含时: 把 $\hat{H}_0$ 中的原 $\omega \rightarrow$ 光频即可.

旋转变近似的 $\begin{cases} \text{近共振条件: } \Delta \ll \omega, \nu; g \ll \omega, \nu \\ \text{大失谐条件: } \Delta \gg \Omega_R, g \end{cases}$ 弱耦合确保 Bloch-Steinberg Shift 可略 $\sim \frac{\Omega_R^2}{\omega}$

Rabi 振荡: $\Delta=0$ 时完美振荡; $\Delta \neq 0$ 时 $\Omega \uparrow$ 不能完全振荡.

$\Delta \gg \Omega_R$ 时不再振荡几乎不跃迁.

共振: $\begin{cases} \Omega_R t = \pi/2 & \pi/2 \text{ 脉冲中 } |e\rangle \rightarrow |g\rangle \\ \Omega_R t = \pi & \pi \text{ 脉冲中 } |e\rangle \rightarrow |g\rangle \end{cases}$

偶极矩的计算: (期望值): $P(t) = e \langle \psi(t) | x | \psi(t) \rangle = e \langle \psi(t) | (|a\rangle\langle a| + |b\rangle\langle b|) x (|a\rangle\langle a| + |b\rangle\langle b|) | \psi(t) \rangle$

以光频 ω 振荡. $= C_a^* C_b g_{ab} + c.c. = \tilde{C}_a^* \tilde{C}_b g_{ab} e^{i\omega t} + c.c. = 2 \text{Re}(\tilde{C}_a^* \tilde{C}_b g_{ab} e^{i\omega t})$

旋转变近似的涉及能量守恒和旋向的选择定则; 合理选偏振可不用 RWA (零级的能级)

密度矩阵表示: $\dot{\rho} = -\frac{i}{\hbar} [\hat{H}, \rho] - \frac{1}{2} \{T, \rho\} \Rightarrow \begin{cases} \dot{\rho}_{aa} = -\gamma_a \rho_{aa} + \frac{i}{\hbar} [g_{ab} E \rho_{ba} - c.c.] \\ \dot{\rho}_{bb} = -\gamma_b \rho_{bb} - \frac{i}{\hbar} [g_{ab} E \rho_{ba} - c.c.] \\ \dot{\rho}_{ab} = \dot{\rho}_{ba}^* = -(i\omega + \gamma_{ab}) \rho_{ab} - \frac{i}{\hbar} g_{ab} E (\rho_{aa} - \rho_{bb}) \end{cases}$

$T = \begin{pmatrix} \gamma_a & 0 \\ 0 & \gamma_b \end{pmatrix}$, $\gamma_{ab} = \frac{1}{2}(\gamma_a + \gamma_b)$

$\dot{\rho}_{ab} = \dot{\rho}_{ba}^* = -(i\omega + \gamma_{ab}) \rho_{ab} - \frac{i}{\hbar} g_{ab} E (\rho_{aa} - \rho_{bb})$



失谐时: $\tilde{c}(t) = e^{-i\frac{\Delta^2}{4\Delta}t}$, 由 $\dot{c}_a = \tilde{c}_a e^{-i\omega a t} \Rightarrow \omega a + \frac{\Delta^2}{4\Delta}$ 即 a.c. Stark 效应

Maxwell-Schrödinger 方程:

只需要在上项的方程中加上(如非和)的驱动项, $\rho_{aa} = \dots + r_a \rho_{aa}^{(0)}$, $\rho_{bb} = \dots + r_b \rho_{bb}^{(0)}$
思路: 演化方程描述在 E 作用下密度矩阵如何演化, 而 ρ 的非对角项对应态到态的跃迁
 ρ 的变化, ρ 反过来通过 Maxwell 方程影响 E, 作用-反作用耦合形成一维自治方程

(慢变条件): $\frac{\partial \rho}{\partial t} \ll \nu \rho$, $\frac{\partial \rho}{\partial z} \ll k \rho$; $\frac{\partial \phi}{\partial t} \ll \nu$, $\frac{\partial \phi}{\partial z} \ll k$ (2π 同化成 1).

含义: 导数的余半 $\times$ 一个周期 $\ll$ 物理量自身的值.

激光理论: 光腔的品质因数 $Q = \omega$. 腔内存储的能量 / 每秒消耗的能量 $= 2\pi \nu T$

T: 寿命, ν : 光腔中的频率 阈值条件: 增益 > 损耗 $A > l$;

$\nu = \frac{\nu_c + \tilde{\omega}}{1 + \tilde{\epsilon}}$ 通常 $\tilde{\epsilon}$ 较小, $\nu \rightarrow \nu_c$, 介于 ν_c 与 ω 之间.

五. 全量子理论: 哈密顿量 $\hat{H} = \hat{H}_F + \hat{H}_A + \hat{H}_{AF}$; $\hat{H}_F$: 自发辐射各种模式; 腔单模少模

一般情形: $\hat{H} = \sum_k \hbar \omega_k \hat{a}_k^\dagger \hat{a}_k + \sum_i \hbar \epsilon_i \sigma_i + \hbar \sum_{i,j,k} g_{jk} \sigma_{ij} (\hat{a}_k + \hat{a}_k^\dagger)$ (其中 $g_{jk} = -\frac{\delta \mathbf{E}_j \cdot \mathbf{E}_k}{\epsilon_0 \hbar}$, $\sigma_{ij} = |i\rangle\langle j|$)

二能级: $\hat{H} = \frac{1}{2} \hbar \omega \hat{\sigma}_z + \sum_k \hbar g_k \hat{a}_k^\dagger \hat{a}_k + \hbar \sum_k g_k (\hat{\sigma}_+ + \hat{\sigma}_-) (\hat{a}_k + \hat{a}_k^\dagger)$ ($g_k = g_k^{ab} = g_k^{ba}$).

RWA $\approx \frac{1}{2} \hbar \omega \hat{\sigma}_z + \sum_k \hbar g_k \hat{a}_k^\dagger \hat{a}_k + \hbar \sum_k g_k (\hat{\sigma}_+ \hat{a}_k + \hat{\sigma}_- \hat{a}_k^\dagger)$ TC.

条件同前: $\sim \omega$ 的误差, $g \ll \omega, \nu$ 才行. 所谓“能量守恒”: 不关心相互作用的条件下.

相互作用哈密顿量中的相互作用哈密顿量:

$\hat{H}_I = \hbar \sum_k g_k (\hat{\sigma}_+ \hat{a}_k e^{i\omega t} + \hat{\sigma}_- \hat{a}_k^\dagger e^{-i\omega t})$

因为单模处 $E = \epsilon \cos \nu t = \frac{\epsilon}{2} (e^{i\nu t} + e^{-i\nu t})$

求解: @ n 率方法: $\Omega_n = \sqrt{\Delta^2 + 4g^2(n+1)}$

$g_k \leftrightarrow -\frac{\Delta_k}{\Sigma} \leftarrow$ (若 $\Sigma=0$)
此时 $\hat{a}$ 中还含有强度信息.

$|b, 0\rangle$ 即使真空态也有跃迁, 衰减没有

$|a, 0\rangle \leftrightarrow |b, 1\rangle$

$|a, 1\rangle \leftrightarrow |b, 2\rangle$

布居: $W(t) = \sum_n |c_n(t)|^2 = \sum_n |c_{b,n}(t)|^2$ 无耗散时完全振荡.

全量子理论的过渡: 半经典理论将原子与光场分开考虑, 给出可完全翻转的预言的准确性:

当相干态 $|\alpha|^2$ 足够大时, 相干态的分布变得极窄, 使得 $\Omega_n t = \pi/2$ 时, 系统的态非常接近

$|g\rangle \otimes |\alpha\rangle \rightarrow |e\rangle \otimes |\alpha\rangle$ (非常像); 有分布的 Fock 分量相差不大近似的完成翻转.

Rabi 振荡周期: $t_R \sim \frac{1}{\Omega_{cn}} = \frac{1}{\sqrt{\Delta^2 + 4g^2cn}}$. ($n \gg 1$).

Collapse-Revival. 性质: 不同步. 崩塌时间 $(\Omega_{cn} + \sqrt{\Delta^2 + 4g^2cn} - \Omega_{cn} - \sqrt{\Delta^2 + 4g^2cn}) t_C \sim 1$.

相干态分布两端相差 π 的相位, 分母用 $f'(\sim) 2\sqrt{\Delta^2 + 4g^2cn}$ 近似, 有 $t_C \sim \frac{1}{2g} \sqrt{1 + \frac{\Delta^2}{4g^2cn}}$

复原时间: $(\Omega_{cn} - \Omega_{cn} - 1) t_R \sim 2\pi m$, $t_R \sim \frac{2\pi m \sqrt{\Delta^2 + 4g^2cn}}{g}$

$\hat{\sigma}_z = 2\hat{\sigma}_+ \hat{\sigma}_- - \hat{1}$, $[\hat{\sigma}_-, \hat{\sigma}_+] = 2\hat{\sigma}_z$, $[\hat{\sigma}_-, \hat{\sigma}_z] = 2\hat{\sigma}_-$, $[\hat{\sigma}_+, \hat{\sigma}_z] = -2\hat{\sigma}_+$ 其它的代入矩阵计算

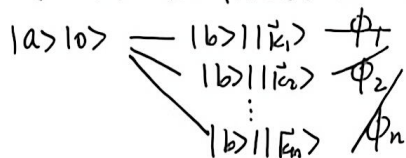
海森堡算符法, 守恒量: $\hat{N} = \hat{a}^\dagger \hat{a} + \hat{\sigma}_+ \hat{\sigma}_-$, $\hat{C} = \frac{1}{2} \Delta \hat{\sigma}_z + g(\hat{\sigma}_+ \hat{a} + \hat{a}^\dagger \hat{\sigma}_-) \Rightarrow [\hat{N}, \hat{H}] = [\hat{C}, \hat{H}] = 0$
适合算时间关联: $\langle \hat{\sigma}_+(t) \hat{\sigma}_-(t') \rangle$, $t \neq t'$ ($\hat{\sigma}_+ \hat{a} + \hat{a}^\dagger \hat{\sigma}_-$) $^{2l+1} = (a a^\dagger)^l a |a\rangle \langle a| + a^\dagger (a a^\dagger)^l |b\rangle \langle b|$

么正算符方法: $(\hat{\sigma}_+ \hat{a} + \hat{a}^\dagger \hat{\sigma}_-)^{2l} = (a a^\dagger)^l |a\rangle \langle a| + (a^\dagger a)^l |b\rangle \langle b|$



• 自发辐射的 Weisskopf - Wigner 理论: 单个二能级原子与多模真空场作用导致衰减

不同模式有区别: 自发辐射是多路分支干涉的结果



长时间 (旋波近似后)

$$\hat{V} = \hbar \sum_{\vec{k}} [g_{\vec{k}}^* (\vec{r}_0) \hat{\sigma}_+ \hat{a}_{\vec{k}} e^{i(\omega - \nu_k)t} + g_{\vec{k}} (\vec{r}_0) \hat{\sigma}_- \hat{a}_{\vec{k}}^\dagger e^{-i(\omega - \nu_k)t}]$$

$g_{\vec{k}}(\vec{r}_0) = g_{\vec{k}} e^{-i\vec{k} \cdot \vec{r}_0}$ 多模时显写出空间相因子, 相位作用会累

$|\psi(t)\rangle = C_a(t)|a, 0\rangle + \sum_{\vec{k}} C_{b,\vec{k}}(t)|b, 1\vec{k}\rangle$, 代入 $i\hbar|\dot{\psi}(t)\rangle = \hat{V}|\psi(t)\rangle$ 有: $C_a(0) = 1, C_{b,\vec{k}}(0) = 0$

$$\begin{cases} \dot{C}_a = -i \sum_{\vec{k}} g_{\vec{k}}^* (\vec{r}_0) e^{i(\omega - \nu_k)t} C_{b,\vec{k}} \\ \dot{C}_{b,\vec{k}} = -i g_{\vec{k}} (\vec{r}_0) e^{-i(\omega - \nu_k)t} C_a \end{cases} \Rightarrow \dot{C}_a = -\sum_{\vec{k}} |g_{\vec{k}}(\vec{r}_0)|^2 \int_0^t dt' e^{i(\omega - \nu_k)(t-t')} C_a(t')$$

代入 $|g_{\vec{k}}(\vec{r}_0)|^2 = \frac{\hbar \nu_k}{2\epsilon_0 V} \frac{1}{\hbar^2} \delta_{ab}^2 \cos^2 \theta$, 求和化积分 $\sum_{\vec{k}} \rightarrow \frac{V}{(2\pi)^3} \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta \int_0^\infty dk k^2$

$$\dot{C}_a = -\frac{4g_{ab}^2}{(2\pi)^2 \hbar \epsilon_0 c^3} \int_0^\infty d\nu_k \nu_k^3 \int_0^t dt' e^{i(\omega - \nu_k)(t-t')} C_a(t')$$

注意边界积分的 1/2!

$$\approx -\frac{\Gamma}{2} C_a(t), \quad \Gamma = \frac{1}{4\pi\epsilon_0} \frac{4\omega^3 \delta_{ab}^2}{3\hbar c^3} \Rightarrow C_a(t) = C_a(0) e^{-\Gamma t/2}, \text{ 则}$$

$$C_{b,\vec{k}}(t) = -i g_{\vec{k}}(\vec{r}_0) \int_0^t dt' e^{-i(\omega - \nu_k)t' - \Gamma t'/2} \Rightarrow C_{b,\vec{k}}(t) = g_{\vec{k}}(\vec{r}_0) \frac{e^{-i\vec{k} \cdot \vec{r}_0}}{(\nu_k - \omega) + i\Gamma/2}$$

$t \gg \Gamma^{-1}$ 的状态: $|0\rangle = \sum_{\vec{k}} g_{\vec{k}} \frac{e^{-i\vec{k} \cdot \vec{r}_0}}{(\nu_k - \omega) + i\Gamma/2} |1\vec{k}\rangle$ 失谐大, 概率小 模平方为洛伦兹型

$\Rightarrow$ 相关性: $G^{(2)}(\vec{r}, \vec{r}; t, t) = \langle \psi | \hat{E}^{(-)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t) | \psi \rangle$, $\hat{E}^{(+)} = \sum_{\vec{k}} \sqrt{\frac{\hbar \nu_k}{2\epsilon_0 V}} \hat{a}_{\vec{k}} e^{-i\nu_k t + i\vec{k} \cdot \vec{r}}$ 有推迟效应.

$\Rightarrow$ 双光子级联的自发辐射: $\hat{V} = \hbar \sum_{\vec{k}} g_{a,b}^* (\vec{r}_0) \hat{\sigma}_+^{(1)} \hat{a}_{\vec{k}} e^{i(\omega_{ab} - \nu_k)t} + \text{H.c.} + \hbar \sum_{\vec{q}} g_{b,c}^* (\vec{r}_0) \hat{\sigma}_+^{(2)} \hat{a}_{\vec{q}} e^{i(\omega_{bc} - \nu_q)t} + \text{H.c.}$ 不考虑的即级联禁戒

$|\psi(t)\rangle = C_a(t)|a, 0\rangle + \sum_{\vec{k}} C_{b,1\vec{k}}(t)|b, 1\vec{k}\rangle + \sum_{\vec{q}} C_{c,1\vec{q}}(t)|c, 1\vec{q}\rangle$ 做类似同理: 下方 $\Gamma_b \uparrow$ 双光子间隔小 改造真空场边界条件, 改造腔内的模式数和模式分布, 将改变(加强或抑制)自发辐射.

Purcell 效应: 腔体共振对准原子频率, 辐射加快 (体积 $V \downarrow$); 失谐率未对准, 辐射被抑制 耦合增强到一定程度比弛豫还强将 Rabi 振荡. $\Gamma_c = \Gamma_0 \propto \lambda^3/V$, Q 为品质因数

$\Rightarrow$ 单原子激光: $|b\rangle|1\vec{k}\rangle \Rightarrow |a\rangle|1\vec{k}\rangle \sim |b\rangle|2\rangle$

• Dark State: 相干囚禁 $\hat{H}_0 = \hbar\omega_a|a\rangle\langle a| + \hbar\omega_b|b\rangle\langle b| + \hbar\omega_c|c\rangle\langle c|$

相干耦合: $\hat{H}_1 = -\frac{\hbar\Omega_{R1}}{2} e^{-i\phi_1} e^{i\Delta_1 t} |a\rangle\langle b| - \frac{\hbar\Omega_{R2}}{2} e^{-i\phi_2} e^{i\Delta_2 t} |a\rangle\langle c| + \text{H.c.}$

令 $C_a = 0$ 可以构造一个只含 $|b\rangle|c\rangle$ 的态, 该态 $C_a(0) = 0, \dot{C}_a = 0$, 则激发

相干布居数囚禁 (CPT) $\Rightarrow$ 态无布居: 暗态较稳定, 亮态有激发成分会通过自发辐射衰减, 稳定在暗态 暗态不吸收光但光与原子体系有较强烈的相互作用, 此时体系的布居能量有变化和光子数反转

相干布居数转移: 绝热演化一段时间使体系始终处于暗态但调节 Ω_{R1}, Ω_{R2} 使态从 $|b\rangle$ 弛豫到 $|c\rangle$ 电磁诱导透明: 强驱动场对弱探测场吸收的影响, 形成暗态后即使共振的探测光

(EIT) 也不被吸收.

核心是多能级耦合的量子相干抑制或消除吸收

无反转激光 (LWI): 通过暗态抑制了 $|b\rangle|c\rangle$ 的受激吸收但 $|a\rangle$ 的受激发射保留, 若有发射 $\rightarrow$ 吸收不需布居反转



微扰论: JC模型在 $(a, n), (b, n+1)$ 子空间的基矢

大失谐有效哈密顿量: $\Delta = \omega - \nu \gg g$: $\hat{V}_{eff} = \hbar \frac{g^2}{\Delta} (\hat{\sigma}_z \hat{a}^\dagger \hat{a} + \hat{\sigma}_+ \hat{\sigma}_-)$ 真空零点能移, 半经典没有 与耦合无关可归入 $\hat{H}_0$

$\omega \uparrow$ $|a\rangle$ $\downarrow$ $|b\rangle$ ω ν Δ $\hat{V}$ $\hat{H}$ $\hat{H} = \frac{1}{2} \hbar \omega \hat{\sigma}_z + \hbar \nu \hat{a}^\dagger \hat{a} + \hbar g (\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger) + \mathcal{E}_0 (\hat{\sigma}_+ e^{-i\nu t} + \hat{\sigma}_- e^{i\nu t})$

量子Raman型: $\hat{H} = \frac{1}{2} \hbar \omega_e |e\rangle\langle e| + \hbar \omega_g |g\rangle\langle g| + \hbar \omega_l |l\rangle\langle l| + \hbar \nu \hat{a}^\dagger \hat{a} + \hbar \nu \hat{b}^\dagger \hat{b} + \hbar g_1 \hat{a}^\dagger |e\rangle\langle e| + \hbar g_2 \hat{b}^\dagger |g\rangle\langle g| + \text{H.c.}$

Raman型大失谐: $\hat{H}_{eff} = \frac{1}{2} \hbar \omega \hat{\sigma}_z + \hbar \nu \hat{a}^\dagger \hat{a} + \hbar \nu \hat{b}^\dagger \hat{b} + \frac{g_1^2}{\Delta} \hat{a}^\dagger \hat{a} |e\rangle\langle e| + \frac{g_2^2}{\Delta} \hat{b}^\dagger \hat{b} |g\rangle\langle g| + \frac{g_1 g_2}{\Delta} (\hat{a}^\dagger \hat{b} \hat{\sigma}_- + \hat{a} \hat{b}^\dagger \hat{\sigma}_+)$

六. 原子光学: de Broglie wave $\lambda = h/p$. 热运动: $\frac{1}{2} m v^2 \sim \frac{n}{2} k_B T$, $\lambda_T \approx \frac{h}{\sqrt{nm k_B T}}$ $n=1,2,3...$ 维度
原子波动性强的条件: $p \downarrow$ 且 $\Delta p \downarrow$ ($p \downarrow \Rightarrow \lambda \uparrow$) 温度低且波矢纯净. 防止形成波包, 又扩散.

原子偏转力: $F_a = \hbar k \cdot \frac{\nu}{t} = \Gamma \rho_{aa} \hbar k$, 在稳态 $\rho_{aa} = 0 = \rho_{bb} = \rho_{ab}$, $\rho_{aa} = \frac{\Omega^2}{4\Delta^2 + \Gamma^2 + 2\Omega^2} < 1/2$
则 $F = \hbar k \Gamma \frac{\Omega^2}{4\Delta^2 + \Gamma^2 + 2\Omega^2}$ 方向与吸收光子束流方向一致.

多普勒冷却: 六束红失谐激光, 迎光力大于背光力 (迎光面失谐减小, 背面失谐增大), $\nu \pm kv$

在饱和效应小 (Ω_R 可略) 的情形下: $F_{\pm} = -2\beta m v$

反冲极限: $\Delta E = \hbar k^2 / 2m$, 多普勒极限: $m \Delta v \sim \frac{\hbar}{2\Delta x}$
 $= \frac{\hbar}{2\Delta t \Delta v} = \frac{\hbar \Gamma}{2\Delta v}$; $\frac{m \langle v^2 \rangle}{2} \sim \frac{3}{2} k_B T$

$$F \approx \hbar k \Gamma \frac{\Omega^2}{4\Delta^2 + \Gamma^2} (1 \pm \frac{\partial \Delta k v}{4\Delta^2 + \Gamma^2})$$

$$F_0 = \hbar k \Gamma \frac{\Omega^2}{4\Delta^2 + \Gamma^2}; \beta = \frac{8\hbar k^2 \Gamma \Omega^2 \Delta}{m(4\Delta^2 + \Gamma^2)^2}$$

七. 耗散: 主方程:

统计涨落, 统计平均, 统计关联. 大热浴 $\hat{\rho}_R \sim \hat{\rho}_R(t_i)$

$$\dot{\rho}_S = -\frac{i}{\hbar} \text{Tr}_R [\hat{V}(t), \rho_S(t_i) \otimes \rho_R(t_i)] - \frac{1}{\hbar} \text{Tr}_R \int_{t_i}^t [\hat{V}(t'), \hat{V}(t''), \rho_S(t'') \otimes \hat{\rho}_R(t'')] dt'$$
 (相互作用绘景)

Markov近似: ① 环境关联时间远小于系统弛豫时间, 从而系统 $\rho_S(t)$ Markov近似

变化还不显著时环境已经忘掉了之前的相互作用, 从而无记忆性.

② 弱耦合: 环境热浴大, 自由度很多, 环境始终保持热平衡不受系统的作用, 从而 ρ_S 的变化率只与当前时刻有关, 与历史无关

a) 热光场热浴: $\hat{V}(t) = \hbar \sum_k g_k (\hat{b}_k^\dagger \hat{\sigma}_- e^{-i(\omega - \nu)k t} + \hat{\sigma}_+ \hat{b}_k e^{i(\omega - \nu)k t})$ 相互作用绘景; 代入计算考作业

$\omega \uparrow$ $|a\rangle$ $\downarrow$ $|b\rangle$ ω ν Δ $\hat{V}$ $\hat{H}$ $\hat{H} = \frac{1}{2} \hbar \omega \hat{\sigma}_z + \hbar \nu \hat{a}^\dagger \hat{a} + \hbar g (\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger) + \mathcal{E}_0 (\hat{\sigma}_+ e^{-i\nu t} + \hat{\sigma}_- e^{i\nu t})$

主方程: $\dot{\rho}_a = -\frac{i}{\hbar} \frac{\Gamma}{2} (\hat{\sigma}_+ \hat{\sigma}_+ \hat{\rho}_a - \hat{\sigma}_+ \hat{\rho}_a \hat{\sigma}_-) - (\bar{n} + 1) \frac{\Gamma}{2} (\hat{\sigma}_+ \hat{\sigma}_- \hat{\rho}_a - \hat{\sigma}_- \hat{\rho}_a \hat{\sigma}_+) + \text{H.c.}$ $\bar{n} = \bar{n}(\omega/c)$

矩阵元的运动方程: 写成矩阵或 $\langle a | \rho | a \rangle = \rho_{aa}$, 求平均 $\rho_{aa} = \rho_{bb} = \rho_{ab} = 0$ 为共振处的值

→ Lindblad方程; b) 场的耗散: 系统为光场 $\hat{a}, \hat{a}^\dagger$, 环境为 $\hat{b}, \hat{b}^\dagger$; 相互作用绘景哈密顿量:

$$\hat{V}(t) = \hbar \sum_k g_k (\hat{b}_k^\dagger \hat{a} e^{-i(\nu - \nu)k t} + \hat{a}^\dagger \hat{b}_k e^{i(\nu - \nu)k t}), \mathcal{E} = \nu/Q, \bar{n} = \bar{n}(\nu/c) \text{ 共振处}; \text{主方程:}$$

$$\dot{\rho} = -\frac{\mathcal{E}}{2} \bar{n} (\hat{a}^\dagger \rho - 2\hat{a} \rho \hat{a}^\dagger + \rho \hat{a} \hat{a}^\dagger) - \frac{\mathcal{E}}{2} (\bar{n} + 1) (\hat{a} \rho \hat{a}^\dagger - 2\hat{a}^\dagger \rho \hat{a} + \rho \hat{a}^\dagger \hat{a})$$
 求运动: $\langle \hat{\sigma}(t) \rangle = \text{Tr}(\hat{\sigma} \rho)$

朗之万方程: $m \dot{u} = -\gamma u + f(t)$, 涨落耗散定理; 单模场在谐振器热浴中的耗散

$\tilde{a}(t) = \hat{a}(t) e^{i\nu t}$, $[\tilde{a}, \tilde{a}^\dagger] = [\hat{a}(t), \hat{a}^\dagger(t)] = 1$, $\dot{\tilde{a}} = -\frac{1}{2} \mathcal{E} \tilde{a} + F_a(t)$, $\mathcal{E} = 2\pi g^2(\nu) X(\nu)$ 共振处值

F_a 与 $\hat{b}_k(0)$ 相关, 满足上述各均值式 $\langle F_a^\dagger(t) \rangle_R = \langle F_a(t) \rangle_R = \langle F_a^\dagger(t) F_a(t') \rangle_R = \langle F_a(t) F_a^\dagger(t') \rangle_R = \frac{\nu^2}{2} \delta(t-t')$
 $\langle F_a^\dagger(t) F_a(t') \rangle_R = \mathcal{E} \bar{n} \delta(t-t')$, $\langle F_a(t) F_a^\dagger(t') \rangle_R = \mathcal{E} (\bar{n} + 1) \delta(t-t')$, $\tilde{a}(t) = \tilde{a}(0) \exp(-\frac{\mathcal{E}}{2} t) + \int_0^t dt' e^{-\frac{\mathcal{E}}{2}(t-t')} F_a(t')$



求 F 和 a 的 $\langle \rangle_R$ 值, 代入 $\tilde{a}(t)$ 或 $\tilde{a}^\dagger(t)$ 表达式, 注意端点处 δ func. 积出来的 $1/2$.

场的运动: 对 Langevin 方程求平均 $\frac{d}{dt} \langle \tilde{a}(t) \rangle_R$; $\frac{d}{dt} \langle \tilde{a}^\dagger(t) \tilde{a}(t) \rangle_R = \langle \frac{d}{dt} \tilde{a}^\dagger(t) \tilde{a}(t) \rangle_R + \langle \tilde{a}^\dagger(t) \frac{d}{dt} \tilde{a}(t) \rangle_R$
代入 Langevin 方程即可 $\Rightarrow$ 9.3.4 m.n 次的运动.

关联的计算: 先解出 $\tilde{a}(t_i + \tau) = \tilde{a}(t_i) \exp(-\frac{\nu}{2Q}\tau) + \int_{t_i}^{t_i+\tau} dt' \exp[-(t_i+\tau-t')(-\frac{\nu}{2Q})] F \tilde{a}(t')$

则 $\langle \tilde{a}^\dagger(t_i) \tilde{a}(t_i + \tau) \rangle_R = \langle \tilde{a}^\dagger(t_i) \tilde{a}(t_i) \rangle_R \exp(-\frac{\nu}{2Q}\tau)$, $\langle \tilde{a}^\dagger(t_i) \tilde{a}(t_i + \tau) \rangle_R = \langle \tilde{a}(t_i) \tilde{a}^\dagger(t_i + \tau) \rangle_R$

对单模场 $G^{(1)}(\vec{r}, \vec{r}; \tau) = \langle \hat{E}^{(+)}(\vec{r}, t) \hat{E}^{(+)}(\vec{r}, t') \rangle$, $t' = t + \tau \Rightarrow G^{(1)}(\vec{r}, \vec{r}; \tau) = \langle \tilde{a}^\dagger(t) \tilde{a}(t + \tau) \rangle e^{-i\nu\tau}$

功率谱: $S(\omega) = \frac{1}{\pi} \text{Re} \int_0^\infty \langle \tilde{a}^\dagger(t) \tilde{a}(t + \tau) \rangle_R e^{i\omega\tau} d\tau = \frac{\langle n \rangle}{\pi} \frac{\nu/2Q}{(\omega - \nu)^2 + (\nu/2Q)^2}$

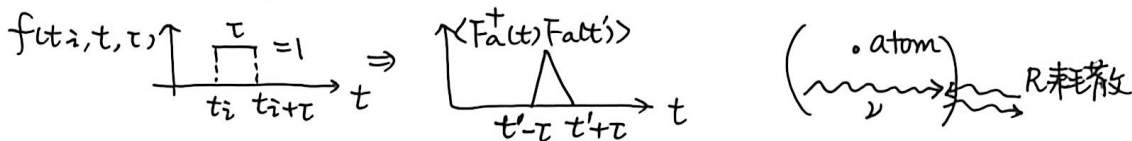
洛伦兹线型: $\omega = \nu$, 全宽度: $2 \cdot \frac{\nu}{2Q} = \frac{\nu}{Q} = \frac{\gamma}{Q}$.

认为功率谱与模式密度呈简单缩放关系, $D(\omega) = \frac{S(\omega)}{\langle n \rangle} = \frac{1}{\pi} \frac{\nu/2Q}{(\omega - \nu)^2 + (\nu/2Q)^2}$.

$\Rightarrow$ 原子热库, 非 Markov 作用: 系统: 光场, 热库: 原子, 有限的相互作用时间.

$\dot{a} = -\frac{1}{2}\gamma a + F a(t)$, $\langle F a(t) F a(t') \rangle = \gamma g^2 (1 + \exp(-\frac{\hbar\nu}{k_B T}))^{-1} \int_{-\infty}^t dt'' f(t'', t, \tau) f(t'', t', \tau)$

并不是只在 $\delta(t-t')$ 有值, 在 $|t-t'| \leq \tau$ 的一段时间内都有关联.



$\Rightarrow$ 耗散真空中的原子: 原子与单模辐射场: Rabi 振荡, 原子与多模(真空)场作用: 自发辐射

原子与修饰的场模式作用: 场耗散研究单模场 $\tilde{a}, \tilde{a}^\dagger$ 在热库 $\hat{b}, \hat{b}^\dagger$ 下的模式密度变化.

原子与光场作用, 光场与热库作用:

自发辐射 $\Gamma = 2\pi \langle |g(\omega)|^2 \rangle D(\omega) \Rightarrow \Gamma_c = 2\pi \langle |g(\omega)|^2 \rangle D_c(\omega)$, 在 $\Delta = 0$ 的近共振条件下:

估计: $\Gamma_c = \Gamma Q (\frac{2\pi c^3}{V \omega^3})$ 自发速率增强 Q 倍.

定量计算: $\hat{\mathcal{H}} = \hat{\mathcal{H}}_F + \hat{\mathcal{H}}_A + \hat{\mathcal{H}}_{AF} + \hat{\mathcal{H}}_R + \hat{\mathcal{H}}_{FR}$ (共振时 $\omega = \nu$)

$= \hbar \omega \hat{a}^\dagger \hat{a} + \frac{1}{2} \hbar \omega \hat{\sigma}_z + \hbar g (\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+) + \sum_k \hbar \nu_k \hat{b}_k^\dagger \hat{b}_k + \hbar \sum_k g_k (\hat{b}_k^\dagger \hat{a} + \hat{a}^\dagger \hat{b}_k)$

求解: $\langle \hat{a}^\dagger \hat{a} \rangle_t$ 和 $\langle \hat{\sigma}_z \rangle_t$, 而 $\rho_a = \text{Tr}(\hat{\rho} \hat{\sigma}_+) = \frac{1}{2} (1 + \langle \hat{\sigma}_z \rangle)$ 为处于上能级的概率.

极限: $\frac{\gamma}{Q} \ll 4g$ 耦合强耗散弱: 主峰 Rabi 振荡带衰减; $\frac{\gamma}{Q} = 2g$ (弱阻尼) $\rho_a \uparrow$
 $\frac{\gamma}{Q} \gg 4g$ 耗散强耦合弱: $\hbar$ 平就是耗散, 来不及振荡, 指数衰减(过阻尼) $\rho_a \uparrow$

$\Rightarrow \Gamma_c = \frac{4g^2}{\gamma} = \Gamma Q (\frac{6\pi c^3}{V \omega^3})$, ($\omega = \nu$ 共振); $Q = \frac{\nu}{\gamma}$



半经典: $\hat{H}_2 = -\vec{d} \cdot \vec{E} = e\vec{r} \cdot \hat{E} \cos \nu t$

($e > 0$) 单个二能级

单模电磁场

$$\vec{d} = \vec{d}_{ab}|a\rangle\langle b| + \vec{d}_{ba}|b\rangle\langle a|$$

$$\vec{d}_{ab} = -e\langle a|\vec{r}|b\rangle, \vec{d}_{ba} = -e\langle b|\vec{r}|a\rangle$$

$$\vec{d}_{ab} \triangleq \vec{d}_{ab} \cdot \hat{E}, \vec{d}_{ba} \triangleq \vec{d}_{ba} \cdot \hat{E}, \vec{d}_{ab} = \vec{d}_{ba}^* = |\vec{d}| e^{i\phi}.$$

$$|\vec{d}| e^{i\phi}$$

$$|\vec{d}| e^{-i\phi}$$

$$\hat{H}_2 = e\vec{r} \cdot \hat{E} \cos \nu t = (\vec{d}_{ab}|a\rangle\langle b| + \vec{d}_{ba}|b\rangle\langle a|) \frac{\mathcal{E}}{2} (e^{i\nu t} + e^{-i\nu t}).$$

旋波近似: $\hat{H}_2 \approx -\frac{\mathcal{E}}{2} (\vec{d}_{ab}|a\rangle\langle b| e^{-i\nu t} + \vec{d}_{ba}|b\rangle\langle a| e^{i\nu t}).$

$$= -\frac{\mathcal{E}}{2} |\vec{d}_{ab}| (\hat{\sigma}_+ e^{-i\nu t} e^{i\phi} + \hat{\sigma}_- e^{i\nu t} e^{-i\phi}).$$

$$\Omega_R = \frac{|\vec{d}_{ab}| \mathcal{E}}{\hbar}, \hat{H}_2 = -\frac{\hbar}{2} \Omega_R (e^{i\phi} e^{-i\nu t} \hat{\sigma}_+ + e^{-i\phi} e^{i\nu t} \hat{\sigma}_-)$$

量子: $\hat{H}_2 = -\vec{d} \cdot \vec{E} = -\vec{d} \cdot (\hat{E}_{\vec{k}} \hat{a}_{\vec{k}} + \hat{E}_{\vec{k}}^\dagger \hat{a}_{\vec{k}}^\dagger)$

$$= -(\vec{d}_{ab}|a\rangle\langle b| + \vec{d}_{ba}|b\rangle\langle a|) \hat{E}_{\vec{k}} (\hat{a}_{\vec{k}} + \hat{a}_{\vec{k}}^\dagger)$$

$$g_{ab} = -\frac{\vec{d}_{ab} \cdot \hat{E}_{\vec{k}}}{\hbar} = -\frac{\vec{d}_{ab} \hat{E}_{\vec{k}}}{\hbar}$$

RWA $\approx \hbar g_{ab} (\hat{\sigma}_+ \hat{a}_{\vec{k}} +$

$$\hbar (g_{ab} \hat{\sigma}_+ \hat{a}_{\vec{k}} + g_{ba} \hat{\sigma}_- \hat{a}_{\vec{k}}^\dagger)$$

设为实数in the case: $\phi=0$

↓

半经典: $\hat{H}_2 = -\frac{\hbar}{2} \Omega_R (\hat{\sigma}_+ e^{-i\nu t} + \hat{\sigma}_- e^{i\nu t}).$

量子: $\hat{H}_2 = \hbar |g_{ab}| (\hat{\sigma}_+ \hat{a}_{\vec{k}} + \hat{\sigma}_- \hat{a}_{\vec{k}}^\dagger)$

$$= \hbar |g_{ab}| (e^{i\phi} \hat{a}_{\vec{k}}^\dagger \hat{\sigma}_+ + e^{-i\phi} \hat{a}_{\vec{k}} \hat{\sigma}_-).$$

由. $\frac{\int \cos \theta d\Omega}{\int d\Omega} = \frac{1}{4\pi} \cdot \frac{4\pi}{3} = 1/3$. 由此 $|\vec{d}_{ba}|^2_{\text{平均}} = |\vec{d}_{ba}|^2/3$. ✓

P207 6.3.10式的 $|\vec{d}_{ab}|^2$ 即定向情况

